SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 2.8: Vectors — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.8: Vectors
Driving Question	DRIVING QUESTION: Why do two matatu journeys of exactly the same distance from Nairobi's CBD end up at completely different destinations — and what mathematical tool do we need to describe a journey completely?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Why Does 8 km Take You to Westlands or South B? Launching the Matatu Phenomenon**	Students observe that two matatu routes departing from Nairobi's CBD can cover exactly 8 km yet arrive at entirely different destinations — one heading northwest to Westlands, another heading south to South B. Teacher presents a simple Nairobi route map and asks students to record what information a passenger would need beyond 'just the distance' to predict where they will end up. Students note that distance alone is insufficient to describe the journey.	Students learn the foundational distinction between scalar and vector quantities. A SCALAR is fully described by magnitude alone (e.g., temperature in Nairobi = 22°C, time to cook ugali = 20 min, the 8 km distance of the matatu journey). A VECTOR requires both magnitude AND direction to be described completely (e.g., '8 km northwest from CBD' uniquely identifies Westlands; '8 km south from CBD' uniquely identifies South B). Students learn that displacement — not distance — is the vector quantity that describes a journey completely. Key vocabulary introduced: magnitude, direction, scalar, vector, displacement.	Teacher uses a large printed or drawn Nairobi CBD map pinned to the board. Mark the CBD as point O. Draw two arrows of equal length — one pointing northwest (labelled 'Route A → Westlands') and one pointing south (labelled 'Route B → South B'). Ask: 'Both arrows are the same length. Why do they point at different landmarks?' (WAIT 30 seconds; cold-call 3 students.) Guide students to articulate that the arrows carry direction information that the number '8 km' alone cannot carry. Introduce the word VECTOR as the mathematical object that captures both pieces of information simultaneously. Pedagogical note: Resist rushing to notation — the goal of Lesson 1 is conceptual anchoring. Students should leave able to explain in their own words why scalars are insufficient for describing journeys; formal column-vector notation is introduced in Lesson 5.
Lesson 2 **Representing Vectors: Arrows, Notation, and Direction on a Bearing**	Students observe multiple drawn arrows on grid paper representing different matatu displacements from Nairobi CBD. Each arrow has a labelled length and a compass	Students learn the conventions for representing vectors: (1) A vector is drawn as a directed line segment (an arrow) where the length represents magnitude and the	Teacher distributes grid paper. Instruct students to draw vector **p** = '5 cm on a bearing of 045°' (northeast) starting from three different points on the page. Ask: 'Are

	bearing or cardinal direction. Teacher presents pairs of arrows and asks students to decide: are these two vectors the same or different? Students discover that vectors are equal only when both magnitude AND direction match — length alone is not enough, and direction alone is not enough.	arrowhead indicates direction. (2) Vectors are named using bold letters (e.g., **a**) or letters with an arrow above (e.g., →AB), where A is the tail (starting point) and B is the tip (end point). (3) Two vectors are EQUAL if and only if they have the same magnitude and the same direction — position on the page does not matter. (4) The NEGATIVE of a vector reverses direction while keeping magnitude. (5) Bearings (three-digit notation, measured clockwise from north) are introduced as a precise way to state direction. Students practise reading and drawing vectors from Nairobi landmarks using compass bearings.	your three arrows the same vector?' (WAIT 20 seconds; cold-call 2 students.) Confirm: yes — same magnitude, same direction, different position → they are equal vectors (free vectors). Then draw a fourth arrow of 5 cm pointing southwest (bearing 225°) and ask: 'How does this relate to **p**?' Guide students to name it −**p**. Pedagogical note: Emphasise the free-vector concept early; students often incorrectly believe two arrows are different vectors simply because they are drawn in different positions on a diagram. Connect back to the matatu context: a matatu can start from Westlands OR from Kenyatta Avenue and still travel the same displacement vector to its destination.
Lesson 3 **Adding Vectors: Finding the Resultant of a Two-Leg Matatu Journey**	Students observe a two-leg matatu journey scenario: a passenger travels from Nairobi CBD to Westlands (vector **a**), then from Westlands to Gigiri (vector **b**). On a grid map, students draw both legs as arrows (tip-to-tail) and then draw a single arrow from the starting point (CBD) directly to the final destination (Gigiri). Students measure this single arrow and compare its length and direction to the two individual legs.	Students learn that the RESULTANT VECTOR (also called the vector sum) is the single vector that produces the same overall displacement as two or more vectors combined. The resultant is found geometrically using the TIP-TO-TAIL method: place the tail of the second vector at the tip of the first; the resultant runs from the tail of the first to the tip of the last. Key properties: (1) Vector addition is commutative — **a** + **b** = **b** + **a** (Triangle Law). (2) The Parallelogram Law provides an equivalent construction when vectors share a common tail. (3) The resultant is NOT simply the arithmetic sum of the magnitudes — direction changes the outcome. Students confirm that the single-journey distance (resultant magnitude) from CBD to Gigiri is less than the total distance travelled (|**a**| + |**b**|) — a real-world insight connecting to triangle inequality.	Teacher models the tip-to-tail construction on the board using the CBD→Westlands→Gigiri scenario on a scaled grid (1 cm = 1 km). Step 1: Draw **a** (CBD to Westlands). Step 2: Place tail of **b** at tip of **a** (Westlands to Gigiri). Step 3: Draw the resultant **r** = **a** + **b** from CBD to Gigiri. Measure |**r**|; compare to |**a**| + |**b**|. Ask: 'If the matatu drove directly from CBD to Gigiri, would it travel the same total distance as via Westlands?' (WAIT 30 seconds; cold-call 3 students.) Then demonstrate commutativity by reversing the order (**b** first, then **a**) and showing the same resultant. Pedagogical note: Students frequently confuse the resultant with the longest side — stress that the resultant joins start to finish, not the longest segment. Use the DQB to record: 'We can now explain WHY two matatus of the same distance end up in different places, AND find where a two-leg journey ends up.'
Lesson 4 **Scaling the Journey: Scalar Multiplication and Subtraction of Vectors**	Students observe what happens to a vector arrow when it is multiplied by different scalars: 2**a**, ½**a**, −1**a**, −2**a**. On grid paper, students draw the original vector **a** representing a matatu displacement and then construct each scaled version,	Students learn that SCALAR MULTIPLICATION scales a vector's magnitude by the absolute value of the scalar while affecting direction based on the sign: (1) If k > 0, the new vector k**a** points in the same direction as **a** with	Teacher uses the matatu context: 'If one matatu completes the journey CBD→Westlands once, what does 2 times that vector mean? What does ½ times mean?' (WAIT 20 seconds; cold-call 2 students.) Draw 2**a** (double length,

	comparing lengths and directions. They record their observations in a table noting magnitude change and direction change for each multiplier.	magnitude k|**a**|. (2) If $k < 0$, the new vector k**a** points in the opposite direction with magnitude |k||**a**|. (3) If $k = 0$, the result is the zero vector. VECTOR SUBTRACTION is defined as adding the negative: **a** − **b** = **a** + (−**b**). Geometrically, −**b** is drawn by reversing the arrowhead of **b**, then added tip-to-tail. Students also learn that PARALLEL VECTORS lie on the same or parallel lines and that any scalar multiple of a vector is parallel to the original — a concept critical for later work on collinearity.	same direction) and ½**a** (half length, same direction) on the board. Then ask: 'What does it mean for a matatu to travel −**a**?' Establish it means the return journey. Model vector subtraction: 'A matatu travels **a** then we subtract **b** — meaning it effectively reverses the **b** leg and adds it.' Construct **a** − **b** geometrically. Pedagogical note: Students often struggle with negative scalar multiplication conflating 'smaller' with 'negative' — use the return matatu journey analogy consistently. Collinearity will resurface in Lesson 6; plant the idea now by asking: 'If 3**a** and **a** are parallel, what does that say about the lines they lie on?'
Lesson 5 **Column Vectors & Position Vectors — Giving Every Nairobi Landmark an Address**	Students observe a coordinate grid overlaid on a simplified Nairobi map with the CBD at the origin. Various landmarks (Westlands, South B, Gigiri, Uhuru Park, Industrial Area) are plotted as points. Students read off the horizontal and vertical components needed to travel from the origin to each landmark and record them as ordered pairs. They compare these component pairs to the arrows drawn in earlier lessons.	Students learn that a COLUMN VECTOR expresses a displacement as a vertical pair of numbers: the top component gives horizontal movement (positive = east/right, negative = west/left) and the bottom component gives vertical movement (positive = north/up, negative = south/down). Written as a 2×1 column matrix, e.g., **a** = (3 above −2) means '3 units east and 2 units south.' A POSITION VECTOR locates a point relative to a fixed origin O; the position vector of point P is written →OP. The displacement vector from point A to point B is →AB = position vector of B minus position vector of A = **b** − **a**. Column vectors are added and subtracted component-by-component. Students convert all previous Nairobi landmark journeys into column vector notation.	Teacher sets up a coordinate grid (x = east–west, y = north–south) with origin at CBD. Plot Westlands at approximately (−3, 2) and South B at (1, −3). Ask: 'Write a column vector for the journey CBD to Westlands and a column vector for CBD to South B. Are they the same?' (WAIT 30 seconds; cold-call 3 students.) Both have magnitude close to $\sqrt{13} \approx 3.6$ km (scale dependent) but completely different column vectors — powerfully demonstrating the driving question answer in algebraic form. Introduce position vectors: →O(Westlands) = (−3 above 2). Then pose: 'If I know position vector of Westlands and position vector of Gigiri, how do I find the displacement vector from Westlands to Gigiri?' Guide students to →(Westlands→Gigiri) = **g** − **w**. Pedagogical note: The column vector representation is the algebraic payoff of the entire unit. Ensure students physically write component-by-component addition — do NOT allow diagonal 'shortcut' arrow addition at this stage without formal justification.
Lesson 6 **Magnitude of a Vector and Collinearity — How Far and Are We on the**	Students observe three matatu stops plotted on a coordinate grid. They are told that Stop A, Stop B, and Stop C might all lie on the same matatu route (the same	Students learn two closely linked concepts: (1) The MAGNITUDE of a column vector (x above y) is calculated using Pythagoras: |**v**| = $\sqrt{(x^2 + y^2)}$, giving the actual straight-	Teacher poses: 'Nairobi Expressway has three toll points. Their position vectors from the CBD are **a** = (2 above 1), **b** = (4 above 2), **c** = (7 above 4). Are they

Same Road?	straight road) or might not. Students calculate the column vectors →AB and →AC and examine whether one is a scalar multiple of the other. They also use a ruler to check whether the three points are visually collinear, then reconcile their visual and algebraic findings.	line distance of the displacement. This links vector notation back to metric distance. (2) Three points A, B, C are COLLINEAR (lie on the same straight line) if and only if →AB = k·→AC for some scalar k — meaning the two vectors are parallel and share point A. Students practise both skills using Nairobi-themed coordinate problems: finding distances between landmarks and determining whether three bus stops lie on the same route. The concept of collinearity consolidates scalar multiplication from Lesson 4 and column vectors from Lesson 5.	collinear — on the same straight road?' (WAIT 45 seconds for student working; cold-call 2 students to share methods.) Calculate →AB = **b** − **a** = (2 above 1) and →AC = **c** − **a** = (5 above 3). Check: is (5 above 3) = k·(2 above 1)? This requires 5 = 2k AND 3 = k — inconsistent, so NOT collinear. Then adjust **c** to (6 above 3) and redo — now →AC = (4 above 2) = 2·(2 above 1) ✓, confirming collinearity. Also compute |→AB| = √(4+1) = √5. Pedagogical note: Students frequently test collinearity by checking only one component ratio. Insist they verify BOTH components give the same scalar k. This rigour pays dividends in examination questions.
Lesson 7 **Translation Vector as a Transformation: Moving Shapes the Matatu Way**	Students observe a triangle drawn on a coordinate grid representing, for example, a triangular plot of land near Uhuru Park. The teacher applies a translation vector (3 above −2), and a new triangle appears on the grid. Students compare the original and image triangles, recording coordinates, side lengths, angles, and orientation. They note what changes and what stays the same after the translation.	Students learn that a TRANSLATION is a geometric transformation that moves every point of an object by the same translation vector **t** = (a above b), so that if a point has coordinates (x, y), its image has coordinates (x + a, y + b). Key properties of translation: (1) Shape and size are preserved (the transformation is an isometry). (2) Orientation is preserved — the image is not flipped or rotated. (3) The object and image are congruent. (4) Every point moves by the same vector — the object 'slides' without turning. Students connect this to the matatu context: every passenger on the matatu undergoes the same displacement vector regardless of where they started seated. Students also practise finding the translation vector given an object and its image by subtracting corresponding coordinate pairs.	Teacher draws triangle PQR with P(1,1), Q(4,1), R(1,4) on the board grid. Apply translation vector **t** = (3 above −2). Ask students to calculate image coordinates before the teacher plots them: P′(4,−1), Q′(7,−1), R′(4,2). (WAIT 40 seconds; cold-call 3 students.) Plot and confirm. Ask: 'What stayed the same? What changed?' (Side lengths ✓, angles ✓, orientation ✓; position ✗.) Then reverse the task: show triangle and its image, ask students to find **t** by computing image-minus-object for one vertex. Pose extension: 'If the matatu translation **t** takes the triangular Jomo Kenyatta Memorial Park footprint to a new position — what single vector undoes this move?' (−**t**.) Pedagogical note: Translation is students' first encounter with vectors as transformations, not just journeys. Emphasise the dual role of the column vector: it is simultaneously a displacement description AND a transformation instruction.
Lesson 8 **Model Build, Consolidation & Final Explanation — Vectors in Nakuru: Synthesising the Complete Journey**	Students revisit the full sequence of matatu journey scenarios from all previous lessons, now set in Nakuru town for a fresh application context. They observe a multi-leg journey: CBD Nakuru → Milimani Estate → Lake Nakuru National Park Gate →	Students confirm and consolidate the full conceptual arc of the sub-strand: (1) Scalars describe magnitude only; vectors describe magnitude AND direction. (2) Vectors are represented as directed arrows or column vectors. (3) Vectors are added	Teacher facilitates a structured gallery-walk model-build: student pairs post their cumulative vector model (started in Lesson 1, revised each lesson) on the wall. Each model should show the Nairobi/Nakuru map with labelled arrows, column vector

	Accommodation Camp. Each leg is given as a column vector. Students draw the complete journey on a coordinate grid, find the overall resultant vector, calculate the straight-line distance from start to camp, and verify whether any three of the stops are collinear.	using the tip-to-tail method or component-by-component addition of column vectors. (4) Scalar multiplication scales magnitude and can reverse direction. (5) The magnitude of a column vector is found using Pythagoras' theorem. (6) Position vectors locate points relative to an origin; displacement vectors →AB = **b** − **a**. (7) Collinearity is confirmed algebraically when →AB = k·→AC. (8) A translation vector moves every point of a shape by the same displacement, preserving congruence. The driving question is answered fully: two journeys of equal distance diverge because distance is a scalar — only by specifying a vector (magnitude + direction, or a column vector) can a journey be described completely.	expressions for each leg, the resultant, magnitude calculation, and a translation example. Teacher circulates with three focus questions: (1) 'Point to where your model shows WHY equal distances lead to different destinations.' (2) 'Show me how your column vectors are added component-by-component.' (3) 'Where does your model show that translation preserves shape?' (WAIT 2 minutes per pair; cold-call 4 pairs to share one insight each.) Close with a whole-class return to the driving question — nominate one student to give the complete explanation aloud (cold-call, WAIT 60 seconds for preparation). Pedagogical note: Assess model completeness against an 8-criterion checklist (one per lesson concept). Reward models that explicitly label the distinction between scalar (distance) and vector (displacement) — this is the heart of the entire sub-strand and the anchor of the driving question.